Mincheol Son

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About

I am a security researcher working on mobile communication systems. My research focuses on LTE, 5G, and 6G security, including (1) testing frameworks that uncover vulnerabilities in real networks and devices, from device basebands to operator core networks, (2) analysis of 3GPP specifications to identify design flaws and ambiguities, and (3) LLM-assisted protocol analysis and security testing. All my work is empirically verified on real devices, using testbeds built with open-source mobile network stacks and software-defined radios. I am also exploring other areas of wireless security, including satellite-based non-terrestrial networks (NTN). My work has been published at ACM CCS, USENIX Security, NDSS, and ACM WiSec, and CITesting received the Distinguished Paper Award at CCS 2025. I have led security research projects with SK Telecom, LG Uplus, and national 5G/6G security programs.

Research Interests

- Cellular Network Security (LTE/5G/6G)
- Baseband Security
- Wireless System Security
- Fuzzing & Testing Frameworks
- LLM-Assisted Protocol Analysis & Security Testing

Education

Ph.D **KAIST (Korean Advanced Institute of Science and Technology)** Sep. 2019 - Aug. 2026

- Electrical Engineering
- Dissertation Title: A Study on Network-Operation-Driven Testing Method for Finding Implementation Vulnerabilities in the Cellular Core Networks
- Advisor: Yongdae Kim

MS **KAIST (Korean Advanced Institute of Science and Technology)** Aug. 2017 - Sep. 2019

- Electrical Engineering
- Thesis Title: Comparative Analysis of Cellular Networks with Fine-grained State Machine
- Advisor: Yongdae Kim

BS **Hanyang University**, Electronic Engineering Mar. 2013 - Aug. 2017

Publications

CITesting: Systematic Testing of Context Integrity Violations in LTE Core Networks 2025
PDF

Mincheol Son, Kwangmin Kim, Beomseok Oh, CheolJun Park, and Yongdae Kim

Distinguished Paper Award

Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security (ACM CCS 2025)

LTESniffer: An Open-source LTE Downlink/Uplink Eavesdropper 2023
PDF

Tuan Dinh Hoang, CheolJun Park, **Mincheol Son**, Taekkyung Oh, Sangwook Bae, Junho Ahn, Beomseok Oh, and Yongdae Kim

Proceedings of the 16th ACM Conference on Security and Privacy in Wireless and Mobile Networks (ACM WiSec 2023)

Preventing SIM Box Fraud Using Device Fingerprinting 2023
PDF
Beomseok Oh, Junho Ahn, Sangwook Bae, **Mincheol Son**, Yonghwa Lee, Min Suk Kang, and Yongdae Kim
Proceedings of the 30th Annual Network and Distributed System Security Symposium (NDSS 2023)

Watching the Watchers: Practical Video Identification Attack in LTE Networks 2022
PDF
Sangwook Bae, **Mincheol Son**, Dongkwan Kim, CheolJun Park, Jiho Lee, Sooel Son, and Yongdae Kim
Proceedings of the 31st USENIX Security Symposium (USENIX Security 2022)

Hiding in Plain Signal: Physical Signal Overshadowing Attack on LTE 2019
PDF
Hojoon Yang, Sangwook Bae, **Mincheol Son**, Hongil Kim, Songmin Kim, and Yongdae Kim
Proceedings of the 28th USENIX Security Symposium (USENIX Security 2019)

Hidden Figures: Comparative Latency Analysis of Cellular Networks with Fine-grained State Machine Models 2019
PDF
Sangwook Bae, **Mincheol Son**, Sooel Son, and Yongdae Kim
Proceedings of the 20th International Workshop on Mobile Computing Systems and Applications (ACM HotMobile 2019)

Awards

Distinguished Paper Award 2025
- ACM Conference on Computer and Communications Security (ACM CCS 2025)
- CITesting: Systematic Testing of Context Integrity Violations in LTE Core Networks

Best Paper Award 2025
- The 26th World Conference on Information Security Applications
- Denial-of-Service Risks in 5G Devices: Empirical Analysis on Connection Restriction Mechanisms

Cybersecurity Paper Competition 2025 2025
- Honorable Mention

Cybersecurity Paper Competition 2021 2021
- First Prize

Patents

KR 10-3016389 (Granted) Sep. 2026
Method and Apparatus for Detecting Context Integrity Violations in Cellular Core Networks

KR 10-2025-0189113 (Filed) Dec. 2025
Method and Apparatus for Memory Crash Detection Based on Network-Side State Control in Wireless Communication System

KR 10-3016382 (Granted) Sep. 2026
Method and System for Generating and Managing Dynamic SUPI to Enhance User Security in 5G

KR 10-3016388 (Granted) Sep. 2026
Dynamic Testing Method and System for Detecting Memory Corruptions in Lower Layers of Cellular Baseband of Mobile Communication Devices

KR 10-2997756 (Granted) Jul. 2026
Malicious User Equipment-Based Security Analysis Method for Cellular Core Networks and the System Thereof

US 12,511,402 (Granted) / KR 10-2514797 (Granted) — Dec. 2025 / Mar. 2023
Security Analysis System and Method Based on Negative Testing for Protocol Implementation of LTE Device

KR 10-2678476 (Granted) — Jun. 2024
Method and Apparatus for Preventing SIM Box Fraud in Mobile Communication Network Using Device Fingerprinting

US 12,010,650 (Granted) / KR 10-2450114 (Granted) — Jun. 2024 / Sep. 2022
FBS Redirection Attack Method Using Unicast Message Injection in LTE and the System Thereof

KR 10-2546948 (Granted) — Jun. 2023
Analysis System for Detection of SIP in VoLTE and the Method Thereof

KR 10-2514809 (Granted) — Mar. 2023
Video Identification Method in LTE Networks and the System Thereof

US 11,405,787 (Granted) / KR 10-2515232 (Granted) — Aug. 2022 / Mar. 2023
Physical Signal Overshadowing Attack Method for LTE Broadcast Message and the System Thereof

US 11,082,866 (Granted) / KR 10-2162024 (Granted) — Aug. 2021 / Sep. 2020
Apparatus and Method for Checking Problem of Mobile Communication Network Using Operational Logic Modeling and Comparative Analysis

Projects

Development of Security Threat Response and Privacy Protection Technologies and International Cooperation for Establishing a Trusted Global Ecosystem in 6G NTN, Institute for Information & Communications Technology Planning & Evaluation (IITP) — 2025 - 2027

Study on Network Security Inspection and Threat Assessment Against Subscriber Information-Based Attacks, SK Telecom Co., Ltd. — 2025

Development of Private 5G Security Technology for Integrated Private 5G and Enterprise Network Security, Institute for Information & Communications Technology Planning & Evaluation (IITP) — 2024 - 2027

Development of Anti-sniffing Technology in Mobile Communication and AirGap Environments, Institute for Information & Communications Technology Planning & Evaluation (IITP) — 2024 - 2026

Automated Security Diagnosis Framework for Cellular Network Protocol using Formal and Comparative Analysis, Institute for Information & Communications Technology Planning & Evaluation (IITP) — 2020 - 2023

Security Testing and Analysis of 5G Non-standalone Networks, LG Uplus Corp. — 2020 - 2021

Intelligent 5G Core Network Abnormal Attack Detection & Countermeasure Technology Development, Institute for Information & Communications Technology Planning & Evaluation (IITP) — 2019 - 2022

A Study on Physical Layer Security for Heterogeneous Wireless Network, Institute for Information & Communications Technology Planning & Evaluation (IITP) — 2018 - 2022

Professional Activities

Secondary Reviewer

IEEE Symposium on Security and Privacy	2021 - 2022, 2024, 2026
USENIX Security Symposium (Security)	2020 - 2026
ACM Conference on Computer and Communications Security (CCS)	2021, 2023, 2025

Network and Distributed System Security Symposium (NDSS)	2020 - 2023, 2025
ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)	2023 - 2025
ACM International Conference on Mobile Computing and Networking (MobiCom)	2021
IEEE European Symposium on Security and Privacy (EuroS&P)	2020